EXAMINATIONS COUNCIL OF ESWATINI

Physical Science

EGCSE

Eswatini General Certificate of Secondary Education

Syllabus

For Examination in 2027-2029

General Changes in the EGCSE Physical Science Syllabus (6884) for 2027-2029
• The 2024 to 2026 syllabus has been revised. • There were some changes which were made in the syllabus. These include: • Content overview section added below Aims • Deletion of some topics e.g carboxylic acids to reduce content • AOC 1 split into two objectives • Some objectives were modified for clarity • Some topics/objectives were reshuffled e.g digital electronics were moved to P14 • Changing some of the command words, 'define' to 'describe'. • Several topics/objectives were unpacked so that it was clear of what would be assessed, for example, use of R_f values in C3.3.2 is not meant to introduce new content but to provide clarity. • All appendices were unpacked such that they are now detailed e. g List of Apparatus You are advised to read the whole syllabus before planning your teaching programme.

Changes in the content	
CHEMISTRY	
Deleted content	**Added content**
• Introduction to chemistry • Hardness of water • Carboxylic acids • natural polymers	• use of R_f values in C3 • Measurement were introduced in C3 • Energy level diagrams in C8 • Classifying acids/alkalis as weak or strong using the pH values and differences in ionisation
PHYSICS	
• Diffraction	• Energy resources in P4.5 • Semiconductors

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ESWATINI GENERAL CERTIFICATE OF SECONDARY EDUCATION

BROAD GUIDELINES

The Ministry of Education and Training is committed, in accordance with the National Policy Statement on Education, to provide a Curriculum and Assessment System so that at the completion of senior secondary education (Form 4 and Form 5), learners will:

- be equipped to meet the changing needs of the Nation, and
- have attained internationally acceptable standards.

Eswatini's National Education Policy Directives

EGCSE syllabuses for studies in Form 4 and Form 5 will individually, and collectively, enable learners to develop **essential skills** and provide a broad **learning experience** which

- inculcates values and attitudes as well as knowledge and understanding,
- encourages respect for human rights and freedom of speech,
- respects the values and beliefs of others, relating to issues of gender, culture and religion,
- develops desirable attitudes and behaviour towards the environment,
- provides insight and understanding of global issues which affect quality of life in Eswatini and elsewhere, e.g., the AIDS pandemic; global warming; maldistribution of wealth; and technological advances.

The National Curriculum for Form 4 and Form 5

Learners will be given opportunities to develop **essential skills** which will overlap across the entire range of subjects studied. These skills are listed below.

- Communication and language skills
- Numeracy skills: mathematical ideas, techniques and applications
- Problem-solving skills
- Technological awareness and applications
- Critical thinking skills
- Work and study skills
- Independent learning
- Collaboration

To develop these skills, learners must be offered **five compulsory subjects** and at least **two elective subjects** chosen from one or more Field of Study.

Compulsory Subjects

- SiSwati – either First Language or Second Language
- English Language
- Mathematics
- Sciences (Biology or Physical Science)
- Religious Education

Fields of Study

- Agriculture Field of Study
- Business Studies Field of Study
- Consumer Science Field of Study
- Social Sciences and Humanities Field of Study
- Technical Field of Study
- Science Field of Study

INTRODUCTION

The Eswatini General Certificate of Secondary Education (EGCSE) syllabuses are designed as two-year courses for examination in Form 5. Physical Science is designed for learners with a wide range of abilities and relevant to those from different backgrounds and experiences. It requires a wide range of learner-centred activities which are based on practical work. This provides learners with opportunities to acquire scientific knowledge and develop skills and processes which will enable them to apply science in everyday situations. As the learners explore and interpret the physical world, emphasis will be directed to the development of innovative ideas, processes and use of scientific equipment in this advancing technological world. It will also prepare candidates for an assessment that will, within familiar and unfamiliar contexts, test expertise, understanding and insight. This syllabus serves as a basis for further studies in science.

All EGCSE syllabuses follow a general pattern. The main sections are:

- Aims
- Assessment Objectives
- Assessment Curriculum Content

Physical Science falls into the Science Compulsory Subjects Group which includes Biology. It is also an Elective Subject in the following Field of Study Groups: Agriculture, Consumer Science and Technical Field of Study.

PRIOR KNOWLEDGE AND SKILLS

Learners beginning this course should normally have completed the Junior Secondary school science or its equivalent. Learners should also have adequate mathematical skills for the content contained in this syllabus.

PROGRESSION

EGCSE Physical Science qualification enables candidates to further their studies at national, regional and international tertiary institutions.

TEACHING HOURS

Appropriate teaching time for the Physical Science syllabus is at least six (6) periods of forty (40) minutes each over a period of sixty weeks/cycles.

TEACHER SUPPORT MATERIAL

A wide range of materials and resources are available to support teachers in Eswatini schools.

The resources include examination reports, syllabuses, past papers and specimen papers are available on ECESWA website www.examscouncil.org.sz.

SPECIAL REQUIREMENTS

Laboratories furnished with functional equipment for conducting practical (see page 10 for the AOC reference, pages 12-13 for the statement about experimental work and the curriculum content and then pages 40 - 49 for the types of activity and the required equipment and apparatus).

AVAILABLE GRADES

Candidates in this syllabus are eligible for Grades A* to G.

AIMS

The aims of the syllabus are the same for all learners and describe the purposes of a course in Physical Science for the EGCSE Examination.

The aims are to:

1. provide, through well designed studies of experimental and practical science, a worthwhile educational experience for all learners, whether or not they go on to study science beyond this level and, in particular, to enable them to acquire sufficient understanding and knowledge to:
   - 1.1 become confident citizens in a technological world, to take or develop an informed interest in matters of scientific import;
   - 1.2 recognise the usefulness, and limitations, of the scientific method and to appreciate its applicability in other disciplines and in everyday life;
   - 1.3 inspire learners to seek, acquire and develop scientific explanations of natural phenomena;
   - 1.4 be suitably prepared for studies beyond the EGCSE level in pure sciences, in applied sciences or in science- dependent vocational courses.
2. develop abilities and skills that:
   - 2.1 develop and enhance scientific knowledge and understanding;
   - 2.2 encourage a systematic approach to problem-solving
   - 2.3 are useful in everyday life and applicable in domestic, environmental and industrial situations;
   - 2.4 are necessary to communicate scientific findings of practical investigations using proper technical scientific terminology;
   - 2.5 encourage efficient and safe practice;
   - 2.6 will evaluate the positive and negative impact of scientific or technological development.
3. develop attitudes relevant to Physical Science such as:
   - 3.1 concern for accuracy and precision;
   - 3.2 objectivity;
   - 3.3 integrity;
   - 3.4 enquiry;
   - 3.5 initiative;
   - 3.6 inventiveness;
   - 3.7 perseverance;
   - 3.8 validity and reliability
4. stimulate learner interest in, and care for, the environment.
5. promote awareness
   - 5.1 of the potential of the indigenous technologies in developing local societies;
   - 5.2 that scientific theories and methods have developed, and continue to do so, as a result of the co-operative activities of groups and individuals;
   - 5.3 that the study and practice of science is subject to social, economic, technological, ethical and cultural influences and limitations;
   - 5.4 that the applications of science may be both beneficial and detrimental to the individual, the community and the environment;
   - 5.5 that science transcends national boundaries and that the language of science, correctly and rigorously applied, is universal.

CONTENT OVERVIEW

The subject content is divided into two sections: Chemistry (C1–C13) and Physics (P1–P15).

Candidates must study both sections.

Chemistry

C1 The particulate nature of matter

C2 Elements, compounds and mixtures

C3 Experimental techniques

C4 Physical and chemical changes

C5 The Periodic Table

C6 Atomic structure and bonding

C7 Stoichiometry

C8 Chemical reactions

C9 Acids, bases and salts

C10 Metals

C11 Electricity and chemistry

C12 Non-metals

C13 Organic chemistry

Physics

P1 Introduction to physics

P2 Speed, velocity and acceleration

P3 Forces

P4 Work, energy and power

P5 Waves

P6 Thermal physics

P7 Electrostatics

P8 Electricity

P9 Electric circuits

P10 Practical electricity

P11 Magnetism

P12 Electromagnetic effects

P13 Atomic physics

P14 Digital electronics

P15 Semi-conductors and LED monitors

ASSESSMENT OBJECTIVES

Assessment Objectives in Physical Science are:

AOA Knowledge with understanding

AOB Handling information and solving problems

AOC Experimental skills and investigations

A description of each Assessment Objective follows.

AOA KNOWLEDGE WITH UNDERSTANDING

Learners should be able to demonstrate knowledge and understanding in relation to:

1. scientific phenomena, facts, laws, definitions, concepts and theories;
2. scientific vocabulary, terminology and conventions (including symbols, quantities and units);
3. scientific instruments and apparatus, including techniques of operation and aspects of safety;
4. scientific quantities and their determination;
5. scientific and technological applications with their social, economic and environmental implications.

The Curriculum Content defines the factual material that candidates may be required to recall and explain. Questions testing this will often begin with one of the following words: define, state, identify, describe, explain (using your knowledge and understanding) or outline. (See Appendix: Glossary of Terms.)

AOB HANDLING INFORMATION AND SOLVING PROBLEMS

Learners should be able, in words or using other written forms of presentation (i.e., symbolic, graphical and numerical), to:

1. locate, select, organise and present information from a variety of sources;
2. translate information from one form to another;
3. manipulate numerical and other data;
4. use information to identify patterns, report trends and draw inferences;
5. present reasoned explanations for phenomena, patterns and relationships;
6. make predictions and hypotheses;
7. solve problems, including some of a quantitative nature.

These Assessment Objectives cannot be precisely specified in the Curriculum Content because questions testing such skills are often based on information which is unfamiliar to the candidate. In answering such questions, candidates are required to use principles and concepts that are within the syllabus and apply them in a logical, deductive manner to a novel situation. Questions testing these objectives will often begin with one of the following words: discuss, predict, suggest, calculate, explain using/why/how or determine. (See Appendix: Glossary of Terms.)

AOC EXPERIMENTAL SKILLS AND INVESTIGATIONS

Learners should be able to:

1. demonstrate knowledge of how to select and safely use techniques, apparatus and materials;
2. follow a sequence of instructions;
3. make and record observations, measurements and estimates;
4. interpret and evaluate experimental observations and data;
5. plan and carry out investigations;
6. evaluate methods and suggest possible improvements (including the selection of techniques, apparatus and materials).

SPECIFICATION GRID

The approximate weightings allocated to each of the Assessment Objectives in the assessment model are summarised in the table below.

Assessment objectives as approximate marks and a percentage of each component

Assessment Objectives	Paper 1 (marks)	Paper 2 (marks)	Papers 3 and 4 (marks)
AOA Knowledge with understanding	23-28	46-51	0
AOB Handling information and solving problems	12-17	29-34	0
AOC Experimental skills and investigations	0	0	40
Weighting in component %	27	53	20

Assessment objectives as a percentage of the qualification

Assessment Objectives	Weighting in EGCSE %
AOA Knowledge with understanding	50 (not more than 25% recall)
AOB Handling information and solving problems	30
AOC Experimental skills and investigations	20
Total	100

Teachers should take note that there is an equal weighting of 50% for skills (including handling information, solving problems, practical, experimental and investigative skills) and for knowledge and understanding. Teacher's schemes of work and the sequence of learning activities should reflect this balance, so that the aims of the syllabus may be met, and the candidates prepared for the assessment.

SCHEME OF ASSESSMENT

All candidates must enter for three papers. These will be Paper 1, Paper 2, and **one** from the practical assessment papers, Paper 3 (Practical Test) or Paper 4 (Alternative to Practical).

There will be a 50:50 weighting of the content between chemistry and physics in each assessment paper.

A description of each paper follows.

The Data Sheet (The Periodic Table of the Elements) will be included in Papers 1 and 2 while Chemistry Practical Notes will be included in Paper 3 and 4.

Paper 1 (1 hour) **Compulsory** short answer paper consisting of 40 marks. The paper will test skills mainly in Assessment Objectives **A** and **B**. This paper will be weighted at 27% of the final total available marks.
Paper 2 (1 hour 15 minutes) Compulsory theory paper consisting of 80 marks of structured questions. The questions will be based on all material and will test skills mainly in Assessment Objectives **A** and **B**. This paper will be weighted at 53% of the final total available marks.
Practical Assessment The purpose of this component is to test appropriate skills in Assessment Objective **C**. Candidates must be entered for one of the following: **Either** **Paper 3** Practical Test (1 hour 15 minutes), consisting of 40 marks, with questions covering experimental skills and investigations. (See Appendix: Assessment Criteria for Practical.) **Or** **Paper 4** Alternative to Practical (1 hour), consisting of 40 marks. This is a written paper designed to test familiarity with laboratory based procedures. (See Appendix: Assessment Criteria for Practical.) The practical assessment will be weighted at 20% of the final total available marks.

EXPERIMENTAL WORK

Experimental work is an essential component of all sciences. Experimental work within science education:

- gives candidates first-hand experience of phenomena;
- enables candidates to acquire practical skills;
- provides candidates with the opportunity to plan and carry out investigations;
- promotes mastery of concepts.

This can be achieved by individual or group experimental work, or by demonstrations which actively involve the candidates.

Throughout the Curriculum Content section of this syllabus some clear indications are given of opportunities to use practical work, using the command words, for example '*describe*' 'perform *experiments* to…' and '*investigate*…' These instructions mean that such statements may be examined in terms of practical skills in Paper 3 or Paper 4, but also in terms of other skills (Assessment Objectives **A** and **B**) in Papers 1 and 2 covering such skills as planning, prediction, recall, explanation, handling data (including calculations) and interpretation of results.

CURRICULUM CONTENT

Notes:

(i) The Curriculum Content is designed to provide guidance to teachers as to what will be assessed in the overall evaluation of the learner. It is not meant to limit, in any way, the teaching programme of any particular school.

(ii) Due to the spiral nature of the curriculum, it is assumed that the elementary concepts of the syllabus have been covered during study at Junior Secondary level.

(iii) The Curriculum Content is set out in topic areas within Chemistry and Physics. The main topic areas and concepts are indicated in **bold**.

(iv) Cross-references are provided to indicate areas of overlap or close association within the syllabus.

(v) It is intended that, in order to avoid difficulties arising out of the use of *l* as the symbol for litre, use of dm^3 in place of l or litre will be made.

Introduction

The syllabus content that follows is divided into **two** sections: Chemistry (C1- C13) and Physics (P1- P15). Candidates must study both sections.

Chemistry is the study of composition of substances and their effect upon one another. Physics is the study of the relationship or interaction between matter and energy.

Physical science is a scientific subject, experimental by nature. Learners in Physical science are expected to use a number of standard laboratory procedures and should be able to name and use apparatus accurately and appropriately, for measuring

- volume (measuring cylinder, burettes, pipettes)
- mass (digital balance, triple beam balance, lever arm balance)
- temperature (laboratory thermometers)
- time (stopwatch and stop clock - analogue and digital)
- potential difference (voltmeter)
- current (ammeter)
- length (metre rule, micrometer screw gauge, vernier calipers)

The following experimental techniques may be found:

- paper chromatography
- methods of purification by use of suitable solvent, filtration, evaporation, crystallisation, distillation (simple and fractional) and sublimation
- determination of purity by melting point and boiling point
- construction of graphs for interpretation of data
- preparation of salts
- preparation and collection of gases
- determination of density
- determination of resistance of wires
- construction of simple circuits
- detection of charges

CHEMISTRY SECTION

C1 Particulate nature of matter

All learners should be able to:

1. state the distinguishing properties of solids, liquids and gases in terms of shape, compressibility, volume
2. describe the structure (including drawings) of solids, liquids and gases in terms of particle separation, arrangement and types of motion
3. describe the states of matter and explain their interconversions in terms of the kinetic particle theory
4. describe the evidence for the movement of particles in gases and liquids i.e. diffusion and Brownian motion
5. describe and explain diffusion in terms of the movement of particles (atoms, molecules or ions)
6. describe and explain dependence of rate of diffusion on molecular mass

C2 Elements, compounds and mixtures

All learners should be able to:

1. define an atom, compound, element, mixture and a molecule
2. demonstrate an understanding of the differences between elements, compounds and mixtures
3. identify elements, compounds and mixtures

C3 Experimental techniques

C3.1 Measurement

All learners should be able to:

1. name and suggest appropriate apparatus for measurement of:
   (a) time – stopwatch/ stop clock
   (b) temperature - laboratory thermometer
   (c) mass – triple beam balance, electronic balance
2. use and describe how to use measuring cylinders, burettes and pipettes to measure the volume of liquids

C3.2 Methods of purification

All learners should be able to:

1. describe methods of purification by use of a suitable solvent:
   (a) filtration
   (b) evaporation
   (c) crystallisation
   (d) distillation
      (i) simple distillation
      (ii) fractional distillation

(e) separating funnel

(f) sublimation

C3.3 Criteria for purity

1. describe paper chromatography
2. interpret simple chromatograms including the use of R_f values
3. describe how chromatography techniques can be applied to colourless substances by exposing chromatograms to locating agents
4. identify substances and test their purity by:
   (a) melting point and boiling point determination
   (b) paper chromatography
5. describe an investigation to determine the melting point and boiling point of pure and impure substances
6. draw and interpret temperature/time graphs as a means of assessing purity
7. interpret a cooling and a heating curve

C4 Physical and chemical change

All learners should be able to:

1. list properties of a:
   (a) chemical change
   (b) physical change
2. identify and describe physical and chemical changes

C5 The Periodic Table

All learners should be able to:

C5.1 Periodic trends

1. describe the Periodic Table as a method of classifying elements and its use in predicting properties of elements
2. explain, for the first 20 elements, the basis of the Periodic Table using the proton number and the simple structure of atoms

   (note that a copy of the Periodic Table will be provided in Papers 1 and 2.)
3. describe the relationship between the periodic number and the number of shells
4. describe the trend from metallic to non-metallic character across a Period
5. identify alkali metals, alkaline earth metals, halogens and noble gases

C5.2 Group properties

1. describe the relationship between group number and the number of outer electrons
2. describe lithium, sodium and potassium in Group I as a collection of relatively soft metals showing a trend in:
   (a) melting point, density
   (b) reaction with water
3. predict the properties of other elements in the Group given data, where appropriate
4. describe chlorine, bromine and iodine in Group VII as a collection of diatomic non-metals showing a trend in colour and state of matter
5. state the reactions of chlorine, bromine and iodine with other halide ions
6. predict the properties of other elements in the Group given data where appropriate
7. identify trends in other groups given information about the elements concerned

C5.3 Transition elements

1. use data to describe and investigate physical properties (density, fixed points, hardness, conductivity and colours of compounds) and chemical properties (variable oxidation states) of the transition metals and their compounds, exemplified by copper and iron
2. state the use of transition elements as catalysts

C5.4 Noble gases

1. describe the noble gases as being unreactive
2. describe the uses of noble gases in providing an inert atmosphere, e.g., argon in lamps; helium for filling weather balloons

C6 Atomic structure and bonding

C6.1 Atomic structure

All learners should be able to:

1. describe the simple structure of atoms in terms of neutrons, protons and electrons
2. state relative charges and approximate relative masses of protons, neutrons and electrons
3. define and use proton (atomic) number and nucleon number
4. define and use nucleon number (mass number) as the total number of protons and neutrons in the nucleus of an atom
5. use proton number and the simple structure of atoms to explain the basis of the Periodic Table, with special reference to the elements of proton numbers 1 to 20
6. deduce information from the notation ${}^{A}_{Z}X$ for an atom
7. describe the build-up of electrons in 'shells' and understand the significance of the noble gas electronic structures and of the outer-shell electrons with special reference to the elements with protons numbers from 1 to 20 (The ideas of the distribution of electrons in s- and p-orbitals and in d-block elements are **not** required.)
8. describe the significance of the outermost electrons and the noble gas electronic configuration

9. define *isotopes* as atoms of the same element which have the same proton number but a different nucleon number
10. understand that isotopes have the same properties because they have the same number of electrons in their outer shell

C6.2 Bonding

C6.2.1 Ions and ionic bonds

1. describe the formation of ions by electron loss or gain
2. describe the formation of ionic bonds between the alkali metals and the halogens
3. describe the formation of ionic bonds between metallic and non-metallic elements to include the strong attraction between ions because of their opposite electrical charges
4. draw 'dot and cross' diagrams to show the formation of ionic bonds
5. describe the lattice structure of ionic compounds as a regular arrangement of alternating positive and negative ions e.g. the structure of sodium chloride

C6.2.2 Molecules and covalent bonds

1. describe the formation of single covalent bonds in H_2, Cl_2, H_2O, CH_4 and HCl as the sharing of pairs of electrons leading to the noble gas configuration
2. draw 'dot and cross' diagrams to show the formation of single covalent bonds
3. describe the electron arrangement in more complex molecules such as N_2, C_2H_4, CH_3OH and CO_2
4. describe and investigate the differences in volatility (including melting point and boiling point), solubility and electrical conductivity between ionic and covalent compounds
5. explain the differences in melting point and boiling point of ionic and covalent compounds in terms of attractive forces

C6.2.3 Macromolecules

1. describe allotropy
2. name the allotropes of carbon as diamond, graphene and graphite
3. describe the structure of
   (a) graphite
   (b) diamond
   (c) graphene
4. relate the structures listed in (C6.2.3.3) to their melting points, conductivities, hardness and uses e.g graphite as a lubricant and a conductor, diamond in cutting tools and graphene as a conductor

C6.2.4 Metallic bonding

1. describe metallic bonding as a lattice of positive ions in a 'sea of electrons'
2. use metallic bonding to describe:
   (a) the electrical conductivity
   (b) malleability of metals

C7 Stoichiometry
All learners should be able to: 1. use the symbols of the elements and write the formulae of simple compounds 2. deduce formulae of simple compounds from relative numbers of atoms present 3. determine the formula of an ionic compound from the charges of the ions present 4. deduce the formula of a simple compound from a model or a diagrammatic representation 5. interpret and balance simple symbol equations 6. construct word equations and simple balanced chemical equations with state symbols, including ionic equations 7. deduce the balanced equation of a chemical reaction given relevant information 8. define *relative atomic mass* (Ar) as the average mass of naturally occurring atoms of an element on a scale where the ^{12}C atom has a mass of exactly 12 units 9. define *relative molecular mass* (Mr) and calculate it as the sum of the relative atomic masses (the term relative formula mass or M_r will be used for ionic compounds) 10. calculate the percentage of mass of components of a compound 11. use the mole and the Avogadro's Constant 12. use molar gas volume taken as 24 dm^3 at room temperature and pressure 13. calculate stoichiometric reacting masses and volumes of gases and solutions, solution concentrations expressed in mol/dm^3 or g/dm^3 (Calculations based on limiting reactants may be examined. Questions on the gas laws and the conversions of gaseous volumes to different temperatures and pressures will **not** be examined.) 14. calculate the empirical formulae of a compound 15. deduce the molecular formulae from the empirical formulae given the molar mass 16. determine limiting reactants in a chemical reaction
C8 Chemical reactions
All learners should be able to: **C8.1 Production of energy** 1. describe the use of hydrogen as a fuel e.g. in rockets 2. describe the use of uranium-235 as a source of energy 3. describe the production of electrical energy from simple cells i.e. two electrodes in an electrolyte (this should be linked with the reactivity series) 4. describe the release of thermal energy by burning fuels

C8.2 Energetics of a reaction

1. describe, using examples, exothermic and endothermic reactions
2. describe bond breaking as endothermic and bond formation as exothermic
3. draw and label energy level diagrams for exothermic and endothermic reactions using data provided
4. Interpret energy level diagrams showing exothermic and endothermic reactions and the activation energy of a reaction
5. describe an experiment to:
    - determine the energy released in combustion of fuels (e.g. ethanol) and foods (e.g. peanuts)
    - perform calculations to find the energy released per unit mass using the formula, $Q = mc\Delta T$

C8.3 Rate (Speed) of a reaction

1. define a catalyst
2. classify catalysts into inorganic and organic (enzymes)
3. investigate and describe the effect on the rate of reactions of:
    - changing concentration
    - changing particle size
    - presence of catalysts (including enzymes) and
    - changing temperature
4. plot graphs and interpret data obtained from experiments concerned with rate of a reaction
5. describe the effect of concentration, particle size, catalysts (including enzyme) and temperature on the rate of reactions in terms of the collision theory
6. describe and explain the effect of concentration in terms of frequency of collisions between reacting particles
7. describe and explain the effect of changing temperature in terms of the frequency of collisions between reacting particles and more colliding particles possessing the minimum energy (activation energy) to react
8. describe the application of the above factors to the danger of explosive combustion with fine powders (e.g. flour mills) and gases (e.g. methane in mines)
9. devise and explain a suitable method for investigating the effect of a given variable on the rate of a reaction which produce a gas

C8.4 Redox

1. describe oxidation and reduction in chemical reactions in terms of:
   (a) oxygen/hydrogen gain/loss
   (b) electron transfer limited to the formation of binary compounds
2. define and identify an oxidising agent as a substance which oxidises another substance during a redox reaction and a reducing agent as a substance which reduces another substance during a redox reaction
3. show awareness that light can provide energy needed for certain chemical reactions by:
   (a) describing the use of silver salts in photography i.e. reduction of silver ions to silver
   (b) stating that photosynthesis leads to the production of glucose from carbon dioxide and water in the presence of chlorophyll and sunlight (energy)

C9 Acids, bases and salts

All learners should be able to:

C9.1 Characteristics and properties of acids and bases

1. define acids and bases in terms of proton transfer, limited to aqueous solutions
2. list common examples of acids and bases
3. classify acids as weak or strong acids using the pH values and differences in ionisation
4. define alkalis as soluble bases
5. classify alkalis as weak or strong alkalis using the pH values and differences in ionisation
6. describe the characteristic properties of acids i.e dilute hydrochloric acid and dilute sulfuric acid as in their reactions with metals, bases, carbonates and their effect on indicators, e.g. litmus paper, Universal indicator, phenolphthalein
7. describe neutrality and relative acidity and alkalinity in terms of pH (whole numbers only) measured using Universal Indicator and pH chart
8. use the ideas of acidity, alkalinity and neutrality to explain acid/base reactions
9. describe and explain applications of neutralisation:
   (a) laboratory preparation of salts
   (b) use of lime to control acidity in soil and water
   (c) use of antacids (e.g. bicarbonate of soda) to control stomach acid

C9.2 Types of oxides

1. classify oxides as either basic or acidic related to metallic and non-metallic character of the element forming the oxide.
2. classify other oxides as neutral or amphoteric given sufficient information

C9.3 Preparation of salts

1. describe and prepare soluble salts from bases, carbonates, metals and ammonium salts

2. prepare, separate and purify insoluble salts (see C3.2 – Methods of purification)

C9.4 Identification of ions

describe and use the following tests to identify:

C9.4.1 Aqueous cations

1. ammonium, calcium, copper(II), iron(II), iron(III) and zinc using aqueous sodium hydroxide and aqueous ammonia as appropriate. (Formulae of complex ions are **not** required)

C9.4.2 Aqueous anions

1. carbonate by reaction with dilute acid and then limewater
2. chloride by reaction under acidic conditions with aqueous silver nitrate
3. iodide by reaction under acidic conditions with aqueous lead(II) nitrate/aqueous silver nitrate
4. nitrate by reduction with aluminium to ammonia
5. sulfate by reaction under acidic conditions with aqueous barium ions

C9.5 Identification of gases

1. identify carbon dioxide using limewater
2. identify hydrogen using a lighted splint
3. identify oxygen using a glowing splint
4. identify ammonia using damp litmus paper
5. identify chlorine using damp litmus paper

C10 Metals

C10.1 Properties

All learners should be able to:

1. compare the general physical properties of metals with those of non-metals with respect to:
   (a) effect of hammering (malleability vs brittleness)
   (b) ductility
   (c) electrical conductivity and heat conductivity
   (d) melting and boiling point
   (e) state at room temperature and pressure

10.2 Reactivity series

1. place in order of reactivity: calcium, aluminium, copper, (hydrogen), iron, magnesium, potassium, sodium, zinc and gold by reference to their reactions, if any, with aqueous ions of other metals, by reference to the reactions, if any, of the elements with:
   (a) water or steam
   (b) dilute hydrochloric acid
   (c) reduction of their oxides with carbon
2. account for the apparent unreactivity of aluminium in terms of the oxide layer adhering to the metal
3. deduce an order of reactivity from a given set of experimental results
4. design experiments to investigate the order of reactivity of metals

C10.3 Extraction of metals

1. describe the ease in obtaining metals from their ores by relating the elements to the reactivity series.
2. name metals that occur native including copper and gold
3. name the main ores of aluminium, copper and iron
4. describe and explain the essential reactions in the extraction of iron in the Blast Furnace

   $C + O_2 \rightarrow CO_2$

   $C + CO_2 \rightarrow 2CO$

   $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
5. outline the manufacture of aluminium from pure aluminium oxide using electrolysis
6. describe the importance of conserving resources
7. state that mining and extraction of metals have significant environmental impacts, limited to deforestation, habitat loss, soil erosion, water pollution and air pollution

C10.4 Uses of metals

1. define an alloy
2. state the composition of elements in the following alloys: brass, bronze, mild steel and stainless steel
3. draw the structural diagrams to show how atoms of other elements can change the properties of the main element in an alloy
4. explain why alloying affects the properties of metals
5. state the important uses of alloys: brass, bronze, mild steel and stainless steel
6. describe the uses of:
   (a) aluminium (electrical cables-good conductor of electricity, aircraft bodies-strength and low density and food containers-resistance to corrosion),
   (b) copper (electrical wiring – good conductor of electricity, cooking utensils-good conductor of heat) related to their properties
7. state the uses of zinc for galvanising and making brass

C11 Electricity and chemistry

All learners should be able to:

1. describe electrolysis as the breakdown of an ionic compound when molten or in aqueous solution by the passage of electricity
2. draw a labelled circuit diagram for an electrolytic cell, and use the terms electrode, electrolyte, anode and cathode
3. describe electrolysis in terms of the ions present and reactions at the electrodes, in terms of gain of electrons by cations and loss of electrons by anions to form atoms
4. describe the electrolysis of molten lead(II) bromide
5. describe the electrolysis of concentrated aqueous sodium chloride (starting materials and essential conditions should be given)
6. describe the electrolysis of dilute sulfuric acid (as essentially the electrolysis of water)
7. predict the likely products and the observations made, using inert electrodes (platinum or carbon), in the electrolysis of a specified binary compound in the molten state or in aqueous solution
8. construct equations for the electrode reactions involved in the manufacture of aluminium, chlorine and sodium hydroxide
9. describe the process of electroplating of metals

C12 Non-metals

All learners should be able to:

C12.1 Air

1. describe the volume composition of air
2. describe the fractional distillation of liquid air to obtain oxygen gas, nitrogen gas and the noble gases for industrial use
3. name common pollutants in air as carbon monoxide, sulfur dioxide, oxides of nitrogen, lead compounds, chlorofluorocarbons (CFCs) and excess carbon dioxide
4. describe the sources of each of the pollutants:
   (a) carbon monoxide from incomplete combustion of carbon-containing compounds
   (b) sulfur dioxide from the combustion of fossil fuels containing sulfur compounds leading to ‘acid’ rain
   (c) oxides of nitrogen from car exhausts
   (d) lead compounds from car exhausts
   (e) excess carbon dioxide from the combustion of fuels and
   (f) CFCs from aerosol sprays
5. state adverse effects of the pollutants on:
   (a) buildings (SO_2 and oxides of nitrogen)
   (b) plants (SO_2 and oxides of nitrogen)

(c) health (oxides of nitrogen, sulfur dioxide, lead compounds, carbon monoxide)

(d) the ozone layer (CFCs)

6. state the composition of catalytic converters in car exhaust systems (palladium, platinum and rhodium)
7. explain the importance of catalytic converters in car exhaust systems to remove carbon monoxide and oxides of nitrogen
8. describe, in outline, how a catalytic converter removes nitrogen monoxide and carbon monoxide from exhaust emissions by reaction over a hot catalyst

   $2CO + O_2 \rightarrow 2CO_2$

   $2NO + 2CO \rightarrow N_2 + 2CO_2$

   $2NO \rightarrow N_2 + O_2$
9. describe the role of carbon dioxide in global warming
10. describe the role of ozone in absorbing ultraviolet (UV) radiation

C12.2 Water

1. describe and perform a chemical test for water using anhydrous copper(II) sulfate and cobalt(II) chloride
2. describe how hydration can be reversed (e.g. by heating hydrated copper(II) sulfate or hydrated cobalt(II) chloride)
3. describe, in outline, the purification of water in terms of filtration, sedimentation and chlorination

C12.3 Hydrogen

1. name the uses of hydrogen in the manufacture of ammonia, margarine and as a fuel in rockets
2. describe the preparation, collection and properties of hydrogen
3. describe formation of hydrogen as a product of electrolysis of water (see C11.6 – Electricity and Chemistry)

C12.4 Oxygen

1. describe the combustion of elements e.g. magnesium
2. describe the properties of oxygen
3. describe the preparation and collection of oxygen using potassium manganate(VII) and hydrogen peroxide
4. state the uses of oxygen including use in oxygen tents, in hospitals and with acetylene in welding
5. describe, in simple terms, respiration, combustion and rusting
6. describe and investigate the conditions necessary for rusting to occur
7. describe methods of rust prevention: paint and other coatings e.g., galvanising to exclude oxygen
8. explain sacrificial protection in terms of the reactivity of zinc and iron

C12.5 Carbon dioxide

1. describe formation of carbon dioxide from:
   (a) the complete combustion of carbon containing substances
   (b) as a product of respiration
   (c) and as a product of the reaction between an acid and a carbonate
2. describe the preparation, collection and properties of carbon dioxide
3. state the uses of carbon dioxide including use in fire extinguishers and fizzy drinks

C12.6 Nitrogen

1. describe the preparation of nitrogen by fractional distillation of liquid air
2. describe the essential conditions in the manufacture of ammonia by the Haber process
3. explain why the conditions used in the manufacture of ammonia are essential to obtaining the best yield of ammonia
4. name the uses of ammonia in the manufacture of fertilisers e.g. ammonium sulfate, ammonium nitrate and in the manufacture of household detergents
5. describe the need for nitrogen, phosphorus and potassium compounds in plant life

C12.7 Carbon and carbonates

1. describe the manufacture of calcium oxide (quick lime) in a kiln from calcium carbonate (limestone) in terms of the chemical reaction involved
2. state some uses of lime and slaked lime in treating acidic soil and neutralising acidic industrial waste products
3. describe the uses of calcium carbonate in the manufacture of iron, glass and cement
4. interpret the ease of decomposition of metal carbonates in terms of the reactivity series

C13 Organic chemistry

All learners should be able to:

C13.1 Name of compounds

1. name, and draw the structure of unbranched alkanes, alkenes, alcohols and acids containing up to six carbon atoms; and the products of the reactions stated in C13.5 –C13.7
2. distinguish between the molecular and structural formula of alkanes, alkenes, alcohols and acids
3. state the type of compound present given a chemical name, ending in *-ane, -ene, -ol* or *-oic acid* or a molecular structure

C13.2 Fuels

1. name fossil fuels as coal, natural gas and petroleum, and state that they produce carbon dioxide on combustion
2. name methane as the main constituent of natural gas
3. describe petroleum as a mixture of hydrocarbons and its separation into useful fractions by fractional distillation

C13.3 Uses of petroleum fractions

1. name the uses of the fractions:
   (a) liquefied petroleum gas, as a fuel for cooking
   (b) petrol in petrol engines
   (c) the paraffin fraction in oil stoves and aircraft fuel
   (d) the diesel fraction for fuel in diesel engines
   (e) the lubricating fraction for lubricants and making waxes and polishes
   (f) bitumen for making roads
2. discuss the hazards associated with the use of petroleum fractions in terms of flammability and harm to the environment

C13.4 Homologous series

1. describe the homologous series as a ‘family’ of similar compounds with similar properties due to the presence of the same functional group
2. describe the general characteristics of a homologous series

C13.5 Alkanes

1 describe the properties of alkanes (exemplified by methane) as being generally unreactive, except in terms of burning
2. predict the structures of longer–chained alkanes given the number of carbon atoms
3. explain physical trends in their density, state of matter, melting and boiling points

13.6 Alkenes

1. describe the catalytic and thermal cracking of alkanes
2. explain why cracking of longer chain alkanes to manufacture alkenes and hydrogen is an important industrial process
3. describe the properties of alkenes in terms of:
   (a) combustion,
   (b) addition reactions with
      (i) bromine
      (ii) hydrogen
      (iii) steam
4. distinguish between saturated and unsaturated hydrocarbons from molecular structures and by simple chemical tests (use of bromine and potassium manganate(VII))

C13.7 Alcohols

1. describe the formation of ethanol by the catalytic addition of steam to ethene
2. describe the formation of ethanol (and carbon dioxide) by fermentation and its importance to the wine and brewing industry
3. describe the properties of alcohols in terms of combustion and dehydration
4. state the uses of ethanol as:
   (a) a solvent
   (b) a fuel
   (c) for sterilisation
   (d) as a constituent of alcoholic beverages
5. state the advantage of using alcohol as a fuel over petrol

C13.8 Macromolecules

1. describe macromolecules (polymers) in terms of large molecules built up from small units (monomers), different macromolecules having different units and/ or different linkages
2. classify macromolecules as man-made/synthetic (poly(ethene), terylene, nylon)
3. state the monomers of synthetic macromolecules (nylon, poly(ethene), and terylene)
4. describe the formation of poly(ethene) as an example of addition polymerisation of monomer units
5. explain why non-biodegradable plastics cause serious pollution problems
6. describe the formation of nylon and terylene macromolecules as examples of condensation polymerisation
7. draw part - structures of the macromolecules (at least four monomer units) in C13.8.2.
8. identify monomers from the structures of given macromolecules with reference to structures in C13.8.7

PHYSICS SECTION

P1 Introduction to Physics

1. All learners should be able to:
2. name quantities and their units including base SI units
3. use and describe how to use metre rules, micrometer screw gauge, Vernier calipers to determine length and volume of regular objects
4. use and describe how to use clocks and other devices for measuring an interval of time including the period of a pendulum
5. obtain an average value for a small distance and for a short interval of time by measuring multiples (including the period of a pendulum)
6. describe and investigate the effect of mass and length on the period of a pendulum
7. use suitable balances to measure mass of solids and liquids
8. measure the volume of regular and irregular objects using the displacement method
9. recall and use the equation $\rho = \frac{m}{V}$
10. determine density of regular and irregular objects
11. determine density of liquids
12. state that an object floats when put in a liquid of higher density than it
13. work with significant figures (see Mathematical requirements on Appendix 4)

P2 Speed, velocity and acceleration

All learners should be able to:

1. define speed and velocity
2. calculate speed/velocity from: speed = total distance ÷ time
3. identify speed as a scalar quantity and velocity as a vector quantity
4. define acceleration
5. calculate the acceleration using:

   $$\frac{\text{change in velocity}}{\text{time taken}}$$
6. recognise from a speed-time graph when a body is
   (a) at rest
   (b) moving with constant speed
   (c) moving with constant acceleration
7. plot and interpret speed-time graphs
8. calculate the area under a speed-time graph to determine the distance travelled for motion with constant acceleration

9. state that the acceleration of free-fall, g, for a body near to the Earth is constant ($g = 10\ m/s^2$)
10. recognise and interpret graphs of motion for which the acceleration is not constant
11. describe qualitatively the motion of bodies falling in a uniform gravitational field with and without air resistance (including reference to terminal velocity)
12. describe some applications on terminal velocity including reference to parachutes and hailstones

P3 Forces

All learners should be able to:

P 3.1 Forces and motion

1. define a force as a push or pull
2. List different types of forces (e.g. gravitational, friction, upthrust, tension, magnetic, electrostatic)
3. describe the ways in which a force may change the motion of a body
4. describe friction as the force between two surfaces in contact which resists motion and results in heating
5. demonstrate an understanding that air resistance is a form of friction
6. state the advantages and disadvantages of friction
7. find the resultant of two or more forces acting along the same line
8. demonstrate an understanding that if there is no resultant force on a body it either remains at rest or continues at constant speed in a straight line
9. use the relationship between force, mass and acceleration ($F = ma$)

P3.2 Mass and weight

1. define *mass* as the measure of the amount of matter in a body
2. demonstrate an understanding that inertia is the property of mass to resist change in motion
3. state that weight is a gravitational force
4. measure the weight of a body using appropriate balances
5. describe, and use the concept of weight as the effect of a gravitational field on a mass
6. distinguish between *mass* and *weight*
7. calculate the weight of a body from its mass ($W = mg$)
8. describe how weights (and hence masses) can be measured and compared using a balance

P3.3 Forces and stretching

1. define the elasticity of an object
2. perform and describe extension-load experiments
3. plot and interpret extension-load graphs
4. state Hooke's law and recall and use the equation, F = kx where k is the spring constant
5. identify and interpret the significance of the term 'limit of proportionality' for an extension-load graph
6. use proportionality in simple calculations

P3.4 Moments

1. describe the moment of a force as a measure of its turning effect and give everyday examples
2. calculate the moment of a force using the product force x perpendicular distance from the pivot, given the necessary information
3. define centre of mass
4. perform and describe an experiment to determine the position of the centre of mass of a plane lamina
5. describe qualitatively, the effect of the position of the centre of mass on the stability of simple objects
6. state the principle of moments
7. perform and describe an experiment (involving vertical forces) to verify that there is no net moment on a body in equilibrium (including calculations)

P4 Work, energy and power

All learners should be able to:

P4.1 Work

1. demonstrate an understanding that work is done against an opposing force
2. recall and use the equation work done = force x distance moved in the direction of force

P4.2 Energy

1. relate energy transfer to work done and state the unit of energy as the joule, J
2. identify different forms of energy including kinetic and potential energy

P4.3 Energy conversion and conservation

1. use the terms kinetic and potential energy in context
2. give and identify examples of energy conversions in kinetic, gravitational potential, chemical potential, elastic (strain), nuclear (fusion and fission) and thermal energy that have occurred as a result of an event or process
3. apply the principle of conservation to simple examples
4. describe energy transfer in terms of work done and make calculations using Work = F x d

5. recall and use the equations k.e.= ½ mv^2 , p.e. = mgh
6. identify examples of energy changes that are less than 100% efficient
7. explain why some energy changes are less than 100% efficient

P4.4 Power

1. define power as energy transferred (work done) per unit time
2. recall and use the equation P = E /t = work/t in simple systems

P 4.5 Energy resources

1. distinguish between renewable and non-renewable sources of energy
2. describe how electricity or other useful forms of energy may be obtained from:
   (a) chemical energy stored in fuel
   (b) energy from water, including the energy stored in waves, in tides, and in water behind hydroelectric dams
   (c) geothermal resources
   (d) nuclear fission
   (e) heat and light from the Sun (solar cells and panels)
   (f) wind energy
3. give advantages and disadvantages of each method in terms of renewability, cost, reliability, scale and environmental impact

P5 Waves

All learners should be able to:

P5.1 Wave properties

1. describe what is meant by wave motion as illustrated by vibrations in ropes, springs and by experiments using water waves
2. name and identify longitudinal and transverse waves; and distinguish between longitudinal and transverse waves
3. define and draw wavefronts
4. state what is meant by wave speed, frequency, wavelength and amplitude
5. identify a crest, trough, amplitude and wavelength in a wave diagram
6. describe reflection, and refraction in water
7. demonstrate the use of water waves to show:
   (a) reflection at a plane surface,
   (b) refraction due to a change of speed
8. recall and use the equation v = f λ

P5.2 Light

1. perform and describe experiments to find the position of an optical image formed by a plane mirror
2. state the characteristics of an optical image formed by a plane mirror
3. perform simple constructions, measurements and calculations to show reflection of light and formation of images by a plane mirror
4. use the law: angle of incidence = angle of reflection
5. describe refraction, including the angle of refraction, in terms of the passage of light through a parallel sided glass block
6. explain *apparent depth* using the refraction of light in water
7. determine and calculate the refractive index using $n = \sin i / \sin r$
8. describe the action of thin lenses (converging and diverging lenses) on light rays
9. perform an experiment to find the focal point and the focal length of a thin converging lens
10. state the characteristics of an optical image formed by a thin converging lens
11. perform simple constructions, measurements and calculations to show refraction of light and formation of images by a thin converging lens
12. use and describe the use of a converging lens as a magnifying glass

P 5.3 Electromagnetic spectrum

1. describe the main features of the electromagnetic spectrum and state that all electromagnetic (e.m.) waves travel at the same speed in a vacuum
2. state the approximate value of the speed of the electromagnetic waves in a vacuum is 3.0×10^8 m / s and is approximately the same in air
3. describe typical properties and uses of radiations in all the different regions of the electromagnetic spectrum including:
   (a) radio, TV, Bluetooth, Wi-Fi, GPS (radio waves)
   (b) satellite television and telephones (microwaves)
   (c) electrical appliances, remote controllers for televisions and intruder alarms (infrared)
   (d) medicine and security (X-rays)
4. state the dangers of exposure to e.m. waves such as in X-rays

P 5.4 Sound

1. state that sound waves are longitudinal
2. state the approximate range of audible frequencies (20 Hz to 20 000 Hz) for a healthy human ear
3. explain why a medium is required for the transmission of sound waves
4. describe the transmission of sound waves in air in terms of compressions and rarefactions
5. relate the loudness and pitch of sound waves to amplitude and frequency
6. describe how the reflection of sound may produce echoes

P6 Thermal Physics

All learners should be able to:

P 6.1 Expansion and contraction

1. describe, using the kinetic theory, the thermal expansion/contraction of solids, liquids and gases
2. explain in terms of intermolecular forces why solids, liquids and gases expand with temperature at different rates
3. identify and describe some of the everyday applications and consequences of thermal expansion/contraction including bimetallic strips in thermostats

P6.2 Thermometry

1. describe how a physical property which varies with temperature may be used for the measurement of temperature and state examples of such properties (volume, potential difference, resistance)
2. describe how to determine the fixed points and use them to calibrate a thermometer
3. describe the structure and function of liquid-in-glass thermometers
4. demonstrate understanding of sensitivity, range and linearity
5. describe the safe disposal of mercury-in-glass thermometers
6. describe the structure and action of a thermocouple and show understanding of its use for measuring high temperatures and those which vary rapidly

P6.3 Change of state

1. describe the difference between boiling and evaporation
2. state the meaning of melting point and boiling point in terms of energy input without change in temperature

P6.4 Thermal energy transfer

P6.4.1 Conduction

1. recognise and name typical good and bad thermal conductors
2. describe experiments to demonstrate the properties of good and bad thermal conductors
3. explain conduction in solids in terms of molecular vibrations and transfer by electrons

P6.4.2 Convection

1. recognise convection as the main method of energy transfer in fluids
2. interpret and describe experiments designed to illustrate convection in liquids and gases (fluids)
3. relate convection in fluids to density changes

P6.4.3 Radiation

1. recognise radiation as the method of energy transfer that does not require a medium to travel through
2. identify infrared radiation as the part of the electromagnetic spectrum often involved in energy transfer by radiation

3. describe the effect of surface colour (black or white) and texture (dull or shiny) on the emission, absorption and reflection of radiation
4. interpret and describe experiments to investigate the properties of good and bad emitters and good and bad absorbers of infrared radiation

P6.4.4 Applications and consequences of heat transfer

1. identify and explain some of the everyday applications and consequences of conduction, convection and radiation

P7 Electrostatics

All learners should be able to:

1. describe simple experiments to show the production of electrostatic charges in terms of electron transfer
2. describe the detection of electrostatic charges
3. state that there are positive and negative charges
4. state that like charges repel and unlike charges attract
5. state that charge is measured in Coulombs
6. explain in simple terms the occurrence of the phenomenon of lightning

P8 Electricity

All learners should be able to:

P8.1 Current and potential difference

1. define current as the rate of flow of charge
2. state that current in metals is due to a flow of electrons
3. distinguish between conventional current flow and electron current flow
4. recall and use the equation $I = Q/t$
5. use and describe the use of ammeters and voltmeters in measuring current and potential difference
6. state that electromotive force (e.m.f.) of a source of electrical energy is measured in volts
7. describe how e.m.f. is defined in terms of energy supplied by a source in driving a unit charge round a complete circuit
8. distinguish between e.m.f. and potential difference

8.2 Resistance

1. state that resistance = p.d.÷ current
2. describe an experiment to determine V/I characteristics for ohmic (metallic) conductors
3. plot and interpret the V/I characteristic graphs for ohmic (metallic) conductors
4. recall and use the equation V = IR
5. recall and use qualitatively the direct proportionality between resistance and the length and the inverse proportionality between resistance and cross-sectional area of a wire (constantan wire)

P9 Electric Circuits

All learners should be able to:

P9.1 Basic circuits

1. draw and name symbols for sources, switches, resistors (fixed and variable), ammeters, voltmeters, bells, fuses, lamps, relays and diodes
2. draw and interpret circuit diagrams containing sources, switches, resistors (fixed and variable), ammeters, voltmeters, bells, fuses, lamps, relays and diodes as rectifiers

P9.2 Resistors in series and parallel

1. state that current is the same at every point in a series circuit
2. state that for a parallel circuit, the current from the source is larger than the current in each branch
3. determine the combined resistance of two or more resistors in series
4. state that the combined resistance of two resistors in parallel is less than either resistor by itself
5. recall and use the fact that the sum of the potential differences across the components in a series circuit is equal to the total p.d. across the source
6. recall and use the fact that the current from the source is the sum of the currents in the separate branches of a parallel circuit
7. calculate the effective resistance of two resistors in parallel
8. recall and use the fact that the p.d. across separate branches of a parallel circuit is equal to p.d across a battery

P10 Practical electricity

All learners should be able to:

1. describe how to wire a three pin-plug
2. describe the uses of electricity in heating, lighting (including lamps in parallel) and motors
3. recall and use the equations $P = IV$, $E = IVt$
4. describe how fluorescent lights produce light
5. state the hazards of:

   (a) damaged insulation

(b) overheating of cables

(c) damp conditions

6. describe the safe disposal of fluorescent lamps

7. describe and explain the use of electrical safety measures, to include:

(a) fuses

(b) double insulations

(c) earthing

(d) switches

P11 Magnetism

All learners should be able to:

Basic magnetism

1. state the properties of magnets
2. describe magnetic induction
3. distinguish between magnetic and non-magnetic materials
4. describe experiments to identify the pattern of field lines round a bar magnet and two bar magnets
5. distinguish between the magnetic properties of soft iron and steel
6. explain magnetism using simple domain theory

P12 Electromagnetic effects

All learners should be able to:

P12.1 Electromagnets

1. state that a current carrying wire has a magnetic field
2. distinguish between the design of permanent magnets and electromagnets
3. state the factors affecting the strength of an electromagnet

P12.2 Electromagnetic induction

1. describe an experiment which shows that a changing magnetic field can induce an e.m.f. in a circuit
2. state the factors affecting the magnitude of the induced e.m.f.
3. show understanding that the direction of an induced e.m.f. opposes the change causing it
4. distinguish between direct current (d.c) and alternating current (a.c)
5. describe a rotating coil generator and the use of slip rings
6. sketch a graph of voltage output against time for a simple a.c. generator

P12.3 Electric Motor

1. state that a current carrying wire in a magnetic field experiences a force
2. state that a current-carrying coil in a magnetic field experiences a turning effect and that the effect is increased by:
   (a) increasing the number of turns on the coil
   (b) increasing the current
   (c) increasing the strength of the magnetic field
3. relate this turning effect to the action of an electric motor including the action of a split-ring commutator
4. describe applications of an a.c. motor including the control of an automated gate, hairdryers, fans

P12.4 Transformer

1. describe the construction of a basic transformer with a soft-iron core, as used for voltage transformations
2. describe the principle of operation of a transformer
3. use the terms step-up and step-down
4. recall and use the equation $(V_p/V_s) = (N_p/N_s)$ (for 100% efficiency)
5. recall and use the equation $V_pI_p = V_sI_s$ (for 100% efficiency)
6. describe energy loss in cables (no calculation)
7. describe the use of the transformer in high voltage transmission of electricity
8. explain why power losses in cables are lower when the voltage is high

P13 Atomic Physics

All learners should be able to:

P13.1 Nuclear atom

1. describe the composition of the nucleus in terms of protons and neutrons
2. use the term proton number (atomic number), Z
3. use the term nucleon number (mass number), A
4. use the term nuclide and nuclide notation ${}^{A}_{Z}X$
5. use and explain the term isotope

P13.2 Radioactive emissions

1. show awareness of the existence of background radiation
2. state that radioactive emissions occur randomly over space and time
3. state the meaning of radioactive decay
4. describe the three types of radioactive emissions
5. describe the detection of α-particles, β-particles and γ–rays using a GM tube and cloud chamber
6. state, for the three types of radioactive emissions:
   (a) their nature
   (b) their relative ionising effect

(c) and their relative penetrating abilities

7. describe their deflection in electric fields and magnetic fields
8. describe radioactive isotopes such as ^{235}U as a source of energy
9. give and explain examples of practical applications of radioactive isotopes
10. describe how radioactive materials are handled, used and stored in a safe way

P13.3 Radioactive decay and half-life

1. use the nuclide notation in equations to show α and β decay
2. use equations to represent changes in composition of the nucleus during radioactive decay
3. define the term *half-life*
4. use the term *half-life* in simple calculations which might involve information in tables or decay curves
5. calculate half-life from data or decay curves, including curves from which background radiation has not been subtracted

P14 Digital electronics

All learners should be able to:

1. explain and use the terms analogue and digital in terms of continuous variation and high/low states
2. describe the action of NOT, AND, OR, NAND and NOR gates
3. recall and use the symbols for logic gates
4. design and interpret simple digital circuits combining several logic gates
5. use truth tables to describe the action of individual gates and simple combinations of gates

15 Semi-conductors and LED monitors

All learners should be able to:

1. describe a semi-conductor
2. describe types of semi-conductors as intrinsic and extrinsic
3. describe how a semiconductor such as silicon works in doping
4. describe the simple structure of an LED (P-N junction)
5. describe the principle of the operation of an LED
6. describe how energy savers (light emitting diodes (LEDs) produce light
7. explain how a light-emitting diode produces a picture in flat screen monitors

GRADE DESCRIPTIONS

The scheme of assessment is intended to encourage positive achievement by all candidates. Grade descriptions are provided to give a general indication of the standards of achievement likely to have been shown by candidates awarded grades. The grade awarded will depend on the extent to which the candidate has met the assessment objectives overall and may conceal weakness in one aspect of the examination that is balanced by above-average performance on some other.

Grade A	Candidate must show mastery of the curriculum
	A candidate should be able to: • relate facts to principles and theories and vice versa; • state why particular techniques are preferred for a procedure or operation; • select and collate information from a number of sources and present it in a clear logical form; • solve problems in situations which may involve a wide range of variables; • process data from a number of sources to identify any patterns or trends; • generate an hypothesis to explain facts, or find facts to support an hypothesis.
Grade C	Candidate must show a high level of competence in the curriculum.
	A candidate should be able to: • link facts to situations not specified in the syllabus; • describe the correct procedure(s) for a multi-stage operation; • select a range of information from a given source and present it in a clear logical form; • identify patterns or trends in given information; • solve problems involving more than one step, but with a limited range of variables; • generate an hypothesis to explain a given set of facts or data.
Grade G	Candidate must show competence in the curriculum.
	A candidate should be able to: • recall facts contained in the syllabus; • indicate the correct procedure for a single operation; • select and present a single piece of information from a given source; • solve a problem involving one step, or more than one step if structured help is given; • identify a pattern or trend where only a minor manipulation of data is needed; • recognise which of two given hypotheses explains a set of facts or data.

APPENDIX 1: ASSESSMENT CRITERIA FOR PRACTICALS

Practical Assessment

Scientific subjects are, by their nature, experimental. It is, accordingly, important that an assessment of a student's knowledge and understanding of Physical Science should contain a component relating to practical work and experimental skills (as identified by Assessment Objective C). In order to accommodate, within EGCSE, differing circumstances - such as the availability of resources - two alternative means of assessing Assessment Objective C objectives are provided, namely, a formal Practical test and a written Alternative to Practical paper, as outlined in the scheme of assessment.

Candidates may be required to do the following:

- carefully follow a sequence of instructions
- describe, explain or comment on experimental arrangements and techniques
- select the most appropriate apparatus or method for a task and justify the choice made
- draw, complete or label diagrams of apparatus
- perform simple arithmetical calculations
- take readings from an appropriate measuring device or from an image of the device (e.g. thermometer, rule,
- protractor, measuring cylinder, ammeter, stop-watch), including:
    - reading analogue and digital scales with accuracy and appropriate precision
    - interpolating between scale divisions when appropriate
    - correcting for zero errors when appropriate
- plan to take a sufficient number and range of measurements, repeating where appropriate to obtain an average value
- describe or explain precautions taken in carrying out a procedure to ensure safety or the accuracy of
- observations and data, including the control of variables and repetition of measurements
- identify key variables and describe how, or explain why, certain variables should be controlled
- record observations systematically, for example in a table, using appropriate units and to a consistent and appropriate degree of precision
- process data, using a calculator where necessary
- present and analyse data graphically, including the use of best-fit lines where appropriate, interpolation and extrapolation, and the determination of a gradient, intercept or intersection
- interpret and evaluate observations and experimental data
- draw an appropriate conclusion, justifying it by reference to the data and using an appropriate explanation
- comment critically on a procedure or point of practical detail, and suggest an appropriate improvement
- evaluate the quality of data, identifying and dealing appropriately with any anomalous results
- identify possible causes of uncertainty, in data or in a conclusion
- make estimates or describe outcomes which demonstrate their familiarity with an experiment, procedure or technique
- plan an experiment or investigation, including making reasoned predictions of expected results and suggesting suitable apparatus and techniques.

Experimental skills tested in Paper 3 Practical Test and Paper 4 Alternative to Practical

CHEMISTRY

Candidates may be asked questions involving the following experimental contexts:

- simple quantitative experiments involving the measurement of volumes and/or masses
- rates (speeds) of reaction
- measurement of temperature based on a thermometer with 1°C graduations and energetics
- problems of an investigatory nature, possibly including suitable organic compounds
- filtration
- electrolysis
- identification of ions and gases
- metals and the reactivity series
- acids, bases, oxides and preparation of salts
- redox reactions and rusting.

(*Notes for Use in Qualitative Analysis*, will be provided in the question paper.)

PHYSICS

Candidates may be asked questions involving the following experimental contexts:

- measurement of physical quantities such as length or volume or force or density
- cooling and heating
- springs and balances
- timing motion or oscillations
- electrical circuits, circuit diagrams and electrical symbols
- optics equipment such as mirrors, prisms and lenses
- procedures using simple apparatus, in situations where the method may not be familiar to the candidate
- use or describe the use of common techniques, apparatus and materials, e.g. ray-tracing equipment or the connection of electric circuits
- explain the manipulation of the apparatus to obtain observations or measurements, e.g.:
  - when determining a derived quantity, such as the extension per unit load for a spring
  - when testing/identifying the relationship between two variables, such as between the p.d. across a wire and its length
  - when comparing physical quantities, such as two masses, using a balancing method.

(Note: The examination will **not** require the use of textbooks nor will candidates need to have access to their own records of laboratory work made during their course; candidates will be expected to carry out the experiments from the instructions given in the paper.)

Candidates may be required to do the following:

1. demonstrate knowledge of how to select and safely use techniques, apparatus and materials (including following a sequence of instructions where appropriate):

1 identify apparatus from diagrams or descriptions

2 draw, complete or label diagrams of apparatus

3 use, or explain the use of, common techniques, apparatus and materials

4 select the most appropriate apparatus or method for the task and justify the choice made

5 describe how the pH of a solution or substance can be tested

6 describe and explain hazards and safety precautions

7 describe and explain techniques used to ensure the accuracy of observations and data

2. plan experiments and investigations:

1 identify the independent variable and dependent variable

2 describe how and explain why variables should be kept constant

3 suggest an appropriate number and range of values for the independent variable

4 suggest the most appropriate apparatus or technique and justify the choice made

5 describe experimental procedures, including a suitable control experiment

6 identify risks and suggest safety precautions

7 describe how to record the results of an experiment

8 describe how to process the results of an experiment to form a conclusion or to evaluate a prediction

9 make reasoned predictions of expected results

3. make and record observations, measurements and estimates:

1 take readings from apparatus (analogue and digital) or from diagrams of apparatus with appropriate precision

2 take sufficient observations or measurements, including repeats and replicates where appropriate

3 record qualitative observations from tests

4 record observations and measurements systematically, for example in a suitable table, to an appropriate degree of precision and using appropriate units

4. interpret and evaluate experimental observations and data:

1 process data, including for use in further calculations or for graph plotting, using a calculator as appropriate

2 present data graphically

3 analyse and interpret observations and data, including data presented graphically

4 use interpolation and extrapolation graphically to determine a gradient or intercept

5 form conclusions justified by reference to observations and data and with appropriate explanation

6 evaluate the quality of observations and data, identifying any anomalous results and taking appropriate action

5. evaluate methods and suggest possible improvements:

1 evaluate experimental arrangements, methods and techniques, including the use of a control

2 identify sources of error

3 suggest possible improvements to the apparatus, experimental arrangements, methods and techniques

Note on taking readings

When approximate volumes are used, e.g. about 2 cm^3, it is expected that candidates will estimate this and not use measuring devices. A measuring instrument should be used to its full precision. Thermometers may be marked in 1 °C intervals but it is often appropriate to interpolate between scale divisions and record a temperature to the nearest 0.0 °C or 0.5 °C. Measurements using a rule require suitable accuracy of recording, such as 15.0 cm rather than 15 cm; the use of millimetres when appropriate should be encouraged. Similarly, when measuring current, it is often more appropriate to use milliamperes rather than amperes.

APPENDIX 2: APPARATUS AND MATERIALS LIST

The list below details the apparatus expected to be generally available for both the teaching and the examination of Paper 3. The list is not exhaustive: in particular, some items that are commonly regarded as standard equipment in a science laboratory are not included.

The *Confidential Instructions*, provided to Centres prior to the examination of Paper 3, will give the detailed requirements for each examination.

List of apparatus

- rulers capable of measuring to 1 mm
- metre rule
- means of writing on glassware (eg markers, stickers)
- beakers, 50 cm^3, 100 cm^3, 250 cm^3
- evaporating dishes/ crucibles
- a polystyrene or other plastic beaker of approximate capacity 150 cm^3
- test-tubes (Pyrex or hard glass), approximately 125 mm × 16 mm
- boiling tubes, approximately 150 mm × 25 mm
- delivery tubes
- conical flasks, within the range 100 cm^3 to 250 cm^3
- measuring cylinders, within the range 10 cm^3 to100 cm^3
- dropping pipettes
- white tiles
- large containers (e.g. plastic bowl) to hold cold water
- thermometers, –10 °C to +110 °C with 1 °C graduations
- stop clocks (digital and analogue), to measure to an accuracy of 0.01s
- glass rods
- spatulas
- wooden splints
- indicators (e.g. litmus paper, Universal Indicator, full range (from 1 to 14) Universal Indicator pH chart)
- common reagents for tests (e.g. limewater test)
- burettes, 50 cm^3, 100 cm^3
- pipettes, 25 cm^3
- pipette fillers
- filter funnels and filter paper
- wash bottle
- mounted needle
- an ammeter 0 -1 A, 0 – 5 A (analogue and digital)
- voltmeter 0 - 5 V (analogue and digital)

- galvanometer
- electrical cells (batteries) and holders to enable several cells to be joined
- connecting leads and crocodile clips
- d.c. power supply, variable to 12 V
- low-voltage filament lamps in holders
- low voltage LEDs
- various fixed resistors
- rheostats
- resistance wire (constantan wire)
- insulated copper wire
- switches
- good supply of masses and holders (10 g, 20 g, 100 g)
- 2cm expendable springs
- clamps and stands
- pendulum bobs
- newton meters (spring balance / force meter)
- plasticine or modelling clay
- thin inextensible strings/ threads/ twine
- polythene rods
- cellulose acetate rods
- dynamics trolleys
- charts/cards
- bar magnets
- iron nails (4 inches)
- wooden boards
- converging/convex lens with f = 15.0cm
- concave/diverging lens
- glass or Perspex block, rectangular and semi-circular
- optics pins/ steel pins
- plane mirrors
- ray box
- ball and ring apparatus
- iron filings

List of Materials

These items should be available for use in the Practical Test. These lists are not exhaustive and we may also require other items to be sourced for specific exams. The Confidential Instructions we send before the Practical Test will give the detailed requirements for the exam.

Every effort is made to limit the resources required by centres and so minimise the costs. Experiments will be designed around basic apparatus and materials which should be available in most school laboratories or are easily obtainable. Candidates must be provided with appropriate safety equipment, such as suitable eye protection and gloves, during practical work.

Hazard codes are used where relevant and in accordance with information provided by CLEAPSS (www.cleapss.org.uk). Candidates should be familiar with the meanings of these codes and terms but will **not** be assessed on them.

C corrosive
HH health hazard
F flammable
N hazardous to the aquatic environment
MH moderate hazard
T acutely toxic
O oxidising

The attention of centres is drawn to any national regulations relating to safety, first aid and disposal of chemicals. 'Hazard Data Sheets' should be available from your chemical supplier.

The Confidential Instructions will indicate which hazard symbols are applicable for the materials required for each Practical Test exam.

Centres are recommended to maintain stocks of all the materials detailed in this section as they are regularly used in the practical exams.

The following table gives guidance for technicians for the preparation of solutions commonly used in practical papers. Candidates are not expected to be familiar with these preparations.

Label	Identity	Instructions
dilute hydrochloric acid	2.0 mol dm^{-3} HCl	Dilute 170 cm^3 of concentrated (35–37%; approximately 11 mol dm^{-3}) hydrochloric acid **[C][MH]** to 1 dm^3.
dilute nitric acid [C]	2.0 mol dm^{-3} HNO_3	Dilute 128 cm^3 of concentrated (70%) nitric acid **[C][O]** to 1 dm^3.
dilute sulfuric acid [MH]	1.0 mol dm^{-3} H_2SO_4	Cautiously pour 55 cm^3 of concentrated (98%) sulfuric acid **[C]** into 500 cm^3 of distilled water with continuous stirring. Make the solution up to 1 dm^3 with distilled water. ***Care:*** *concentrated H_2SO_4 is very corrosive*.
aqueous ammonia [C][MH][N]	2.0 mol dm^{-3} NH_3	Dilute 112 cm^3 of concentrated (35%) ammonia **[C][MH][N]** to 1 dm^3.

aqueous sodium hydroxide [C]	1.0 mol dm^{-3} NaOH	Dissolve 40.0 g of NaOH **[C]** in each dm^3 of solution. ***Care***: *the process of solution is exothermic and any concentrated solution is very corrosive.*
barium chloride or barium nitrate	0.1 mol dm^{-3} $BaCl_2$ or 0.1 mol dm^{-3} $Ba(NO_3)_2$	Dissolve 24.4 g of $BaCl_2$•$2H_2O$ **[T]** in each dm^3 of solution or dissolve 26.1 g of $Ba(NO_3)_2$ **[HH][O]** in each dm^3 of solution.
silver nitrate [N]	0.05 mol dm^{-3} $AgNO_3$	Dissolve 8.5 g of $AgNO_3$ **[C][O][N]** in each dm^3 of solution.
limewater [MH]	saturated aqueous $Ca(OH)_2$	Prepare fresh limewater by leaving distilled water to stand over solid calcium hydroxide **[C][MH]** for several days, shaking occasionally. Decant or filter the solution.

Label	**Identity**	**Instructions**
acidified aqueous potassium manganate(VII) [MH]	0.01 mol dm^{-3} $KMnO_4$ in 0.5 mol dm^{-3} H_2SO_4	Mix equal volumes of 0.02 mol dm^{-3} $KMnO_4$ and 1.0 mol dm^{-3} H_2SO_4 **[MH]**.
starch solution	freshly prepared aqueous starch solution (approx 2% solution w/v)	Mix 2 g of soluble starch with a little cold water until a smooth paste is obtained. Add 100 cm^3 boiling water and stir. Boil until a clear solution is obtained (about 5 minutes).
methyl orange indicator [F][HH][MH]	methyl orange indicator (pH range 2.9 to 4.6)	Use commercially produced solution or dissolve 0.4 g of solid indicator **[T]** in 200 cm^3 of ethanol (IMS) **[F][HH][MH]** and make up to 1 dm^3 with distilled water.
bromophenol blue indicator [F][HH][MH]	bromophenol blue indicator (pH range 3.0 to 4.5)	Dissolve 0.4 g of the solid indicator in 200 cm^3 of ethanol (IMS) **[F][HH][MH]** and make up to 1 dm^3 with distilled water.
thymol blue indicator [F][HH][MH]	thymol blue indicator (pH range 8.0 to 9.6)	Dissolve 0.4 g of the solid indicator in 200 cm^3 of ethanol (IMS) **[F][HH][MH]** and make up to 1 dm^3 with distilled water.
thymolphthalein indicator [F][HH][MH]	thymolphthalein indicator (pH range 9.3 to 10.5)	Dissolve 2.0 g of the solid indicator in 1 dm^3 of ethanol (IMS) **[F][HH][MH]**.

In addition to the materials in the table, the following materials are recommended to be used in the centre as part of the practical course:

- hydrogen peroxide
- aqueous iodine (approximately 0.01 mol dm^{-3} in 0.2 mol dm^{-3} potassium iodide)
- iron(II) sulfate or ammonium iron(II) sulfate
- copper(II) sulfate
- iron, magnesium, copper and zinc metals
- potassium iodide
- sodium thiosulfate
- solid hydrated barium chloride and magnesium sulfate
- calcium carbonate, sodium carbonate or sodium hydrogencarbonate
- magnesium ribbon
- pH indicator papers
- universal indicator (paper or solution)
- the carbonates (where they exist), sulfates, nitrates and chlorides of the cations listed in the Qualitative analysis notes on page 54
- the sodium and/or potassium salts of the anions listed in the Qualitative analysis notes on page 54
- ethanol
- ethanoic acid
- hydrochloric acid
- magnesium oxide
- aluminium oxide
- zinc oxide
- ethanol
- lead(II) nitrate
- anhydrous copper(II) sulfate
- anhydrous cobalt(II) chloride
- ethanoic acid
- aluminium foil

Note: the standard concentration for most laboratory stock solutions is usually 1 M.

These materials should be kept in stock as they may be required for the practical examination.

APPENDIX 3: SYMBOLS AND UNITS

Learners should be able to state the symbols for the following physical quantities and, where indicated, state the units in which they are measured.

Litre/dm³: to avoid any confusion over the symbol for litre (l or litre), dm^3 will always be used.

Quantity	***Symbol***	***Unit***
wavelength	λ	mm, cm, m
length	*l, h ...*	km, m, cm, mm
area	*A*	m^2, cm^2
volume	*V*	m^3, dm^3, cm^3
weight	*W*	N
mass	*m*	kg, g, mg
density	*d, ρ*	kg/m^3, g/cm^3
time	*t*	h, min, s, ms
speed	*u, v*	km/h, m/s, cm/s
acceleration	*a*	m/s^2
acceleration of free fall	*g*	m/s^2
Gravitational field strength	*g*	*N/kg*
force	*F*	N
Moment of a force		Nm
work done	*w, E*	J, kJ, MJ
energy	*E*	J, kJ, MJ
power	*P*	W, kW, MW
pressure	*p*	N/m^2, Pa
temperature	*θ, T*	°C
focal length	*f*	cm, mm
angle of incidence	*i*	degree (°)
angle of reflection/refraction	*r*	degree (°)
potential difference/voltage	*V*	V, mV
current	*I*	A, mA
e.m.f.	*Ɛ*	V
resistance	*R*	Ω
frequency	*f*	Hz, kHz
wavelength	*λ*	m, cm
focal length	*f*	cm
angle of incidence	*i*	degree (°)
angle of reflection, refraction	*r*	degree (°)
refractive index	*n*	

APPENDIX 4: MATHEMATICAL REQUIREMENTS

Calculators may be used in all parts of the Assessment. Candidates should be able to:

1. add, subtract, multiply and divide
2. understand and use averages, decimals, fractions, percentages, ratios and reciprocals
3. recognise and use standard notation, including both positive and negative indices
4. understand significant figures and use them appropriately
5. recognise and use direct and inverse proportion
6. use positive, whole number indices in algebraic expressions
7. draw charts and graphs from given data
8. interpret charts and graphs
9. determine the gradient and intercept of a graph
10. select suitable scales and axes for graphs
11. recognise and use the relationship between length, surface area and volume and their units on metric scales
12. make approximate evaluations of numerical expressions
13. recall and use equations for the areas of a rectangle, triangle and circle and the volumes of a rectangular block and a cylinder
14. use mathematical instruments (ruler, compasses, protractor and set square)
15. understand the meaning of angle, curve, circle, radius, diameter, circumference, square, parallelogram, rectangle and diagonal
16. solve equations of the form $x = y + z$ and $x = yz$ for any one term when the other two are known
17. recognise and use clockwise and anticlockwise directions
18. recognise and use points of the compass (N, S, E, W)
19. use sines and inverse signs

Presentation of data

The solidus (/) is to be used for separating the quantity and the unit in tables, graphs and charts, e.g. time / s for time in seconds.

(a) Tables

- Each column of a table should be headed with the physical quantity and the appropriate unit, e.g. time / s.
- The column headings of the table can then be directly transferred to the axes of a constructed graph.

(b) Graphs

- Unless instructed otherwise, the independent variable should be plotted on the x-axis (horizontal axis) and the dependent variable plotted on the y-axis (vertical axis).
- Each axis should be labelled with the physical quantity and the appropriate unit, e.g. time / s.

- The scales for the axes should allow more than half of the graph grid to be used in both directions, and
- be based on sensible ratios, e.g. 2 cm on the graph grid representing 1, 2 or 5 units of the variable.
- The graph is the whole diagrammatic presentation, including the best-fit line when appropriate. It may have one or more sets of data plotted on it.
- Points on the graph should be clearly marked as crosses (×) or encircled dots (⊙).
- Large 'dots' are penalised. Each data point should be plotted to an accuracy of better than one half of each of the smallest squares on the grid.
- A best-fit line (trend line) should be a single, thin, smooth straight-line or curve. The line does not need to coincide exactly with any of the points; where there is scatter evident in the data, examiners would expect a roughly even distribution of points either side of the line over its entire length. Points that are clearly anomalous should be ignored when drawing the best-fit line.
- The gradient of a straight line should be taken using a triangle whose hypotenuse extends over at least half of the length of the best-fit line, and this triangle should be marked on the graph.

(c) Numerical results

- Data should be recorded so as to reflect the precision of the measuring instrument.
- The number of significant figures given for calculated quantities should be appropriate to the least number of significant figures in the raw data used.

(d) Pie charts

- These should be drawn with the sectors in rank order, largest first, beginning at 'noon' and proceeding clockwise. Pie charts should preferably contain no more than six sectors.

(e) Bar charts

- These should be drawn when one of the variables is not numerical. They should be made up of narrow blocks of equal width that do not touch.

(f) Histograms

- These are drawn when plotting frequency graphs with continuous data. The blocks should be drawn in order of increasing or decreasing magnitude and they should touch.

Conventions (e.g. signs, symbols, terminology and nomenclature)

Syllabuses and question papers conform with generally accepted international practice.

Litre/dm^3

To avoid any confusion concerning the symbol for litre, dm^3 will be used in place of *l* or litre.

Decimal markers

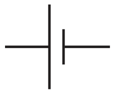
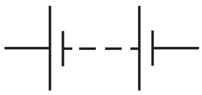
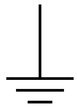
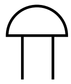
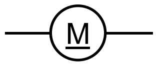
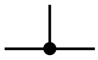
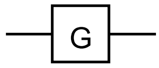
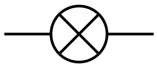
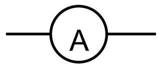
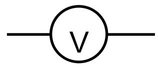
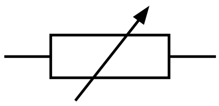
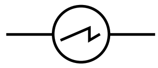
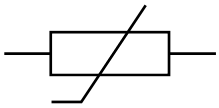
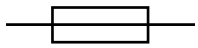
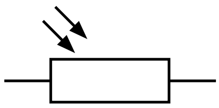
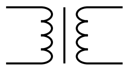
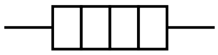
Decimal markers in examination papers will be a single dot on the line. Candidates are expected to follow this convention in their answers.

Numbers

Numbers from 1000 to 9999 will be printed without commas or spaces. Numbers greater than or equal to

10 000 will be printed without commas. A space will be left between each group of three whole numbers, e.g. 4 256 789.

APPENDIX 5: ELECTRICAL SYMBOLS

cell		switch	
battery of cells			
or		earth or ground	
power supply		electric bell	
a.c. power supply		motor	
junction of generator		conductors	
lamp		ammeter	
fixed resistor		voltmeter	
variable resistor		oscilloscope	
thermistor		fuse	
light dependent resistor		transformer	
heater			

APPENDIX 6: NOTES FOR USE IN QUALITATIVE ANALYSIS

Tests for anions

anion	*test*	*test result*
carbonate (CO_3^{2-})	add dilute acid	effervescence, carbon dioxide produced
chloride (Cl^-) [in solution]	acidify with dilute nitric acid, then add aqueous silver nitrate	white ppt.
Iodide ion (I^-) in solution]	Acidify with dilute nitric acid, and then add aqueous lead(II) nitrate/aqueous silver nitrate	yellow ppt.
nitrate (NO_3^-) [in solution]	add aqueous sodium hydroxide, then aluminium foil; warm carefully	ammonia produced
sulfate (SO_4^{2-}) [in solution]	acidify with dilute nitric acid, then add aqueous barium nitrate	white ppt.

Tests for aqueous cations

cation	*effect of aqueous sodium hydroxide*	*effect of aqueous ammonia*
ammonium (NH_4^+)	ammonia produced on warming	-
Calcium (Ca^{2+})	white ppt, insoluble in excess	no ppt or very slightly white ppt.
copper(II) (Cu^{2+})	light blue ppt., insoluble in excess	light blue ppt., soluble in excess, giving a dark blue solution
iron(II) (Fe^{2+})	green ppt., insoluble in excess	green ppt., insoluble in excess
iron(III) (Fe^{3+})	red-brown ppt., insoluble in excess	red-brown ppt., insoluble in excess
zinc (Zn^{2+})	white ppt., soluble in excess, giving a colourless solution	white ppt., soluble in excess, giving a colourless solution

Tests for gases

gas	*test and test result*
ammonia (NH_3)	turns damp red litmus paper blue
carbon dioxide (CO_2)	turns lime water milky
chlorine (Cl_2)	bleaches damp litmus paper
hydrogen (H_2)	'pops' with a lighted splint
oxygen (O_2)	relights a glowing splint

APPENDIX 7: THE PERIODIC TABLE OF THE ELEMENTS

DATA SHEET
The Periodic Table of the Elements

Group I	II											III	IV	V	VI	VII	0
							1 **H** Hydrogen 1										4 **He** Helium 2
7 **Li** Lithium 3	9 **Be** Beryllium 4											11 **B** Boron 5	12 **C** Carbon 6	14 **N** Nitrogen 7	16 **O** Oxygen 8	19 **F** Fluorine 9	20 **Ne** Neon 10
23 **Na** Sodium 11	24 **Mg** Magnesium 12											27 **Al** Aluminium 13	28 **Si** Silicon 14	31 **P** Phosphorus 15	32 **S** Sulfur 16	35.5 **Cl** Chlorine 17	40 **Ar** Argon 18
39 **K** Potassium 19	40 **Ca** Calcium 20	45 **Sc** Scandium 21	48 **Ti** Titanium 22	51 **V** Vanadium 23	52 **Cr** Chromium 24	55 **Mn** Manganese 25	56 **Fe** Iron 26	59 **Co** Cobalt 27	59 **Ni** Nickel 28	64 **Cu** Copper 29	65 **Zn** Zinc 30	70 **Ga** Gallium 31	73 **Ge** Germanium 32	75 **As** Arsenic 33	79 **Se** Selenium 34	80 **Br** Bromine 35	84 **Kr** Krypton 36
85 **Rb** Rubidium 37	88 **Sr** Strontium 38	89 **Y** Yttrium 39	91 **Zr** Zirconium 40	93 **Nb** Niobium 41	96 **Mo** Molybdenum 42	**Tc** Technetium 43	101 **Ru** Ruthenium 44	103 **Rh** Rhodium 45	106 **Pd** Palladium 46	108 **Ag** Silver 47	112 **Cd** Cadmium 48	115 **In** Indium 49	119 **Sn** Tin 50	122 **Sb** Antimony 51	128 **Te** Tellurium 52	127 **I** Iodine 53	131 **Xe** Xenon 54
133 **Cs** Caesium 55	137 **Ba** Barium 56	139 **La** Lanthanum 57 *	178 **Hf** Hafnium 72	181 **Ta** Tantalum 73	184 **W** Tungsten 74	186 **Re** Rhenium 75	190 **Os** Osmium 76	192 **Ir** Iridium 77	195 **Pt** Platinum 78	197 **Au** Gold 79	201 **Hg** Mercury 80	204 **Tl** Thallium 81	207 **Pb** Lead 82	209 **Bi** Bismuth 83	209 **Po** Polonium 84	210 **At** Astatine 85	222 **Rn** Radon 86
223 **Fr** Francium 87	226 **Ra** Radium 88	227 **Ac** Actinium 89 †															

* 58–71 Lanthanoid series
† 90–103 Actinoid series

140 **Ce** Cerium 58	141 **Pr** Praseodymium 59	144 **Nd** Neodymium 60	147 **Pm** Promethium 61	150 **Sm** Samarium 62	152 **Eu** Europium 63	157 **Gd** Gadolinium 64	159 **Tb** Terbium 65	163 **Dy** Dysprosium 66	165 **Ho** Holmium 67	167 **Er** Erbium 68	169 **Tm** Thulium 69	173 **Yb** Ytterbium 70	175 **Lu** Lutetium 71
232 **Th** Thorium 90	231 **Pa** Protactinium 91	238 **U** Uranium 92	237 **Np** Neptunium 93	244 **Pu** Plutonium 94	243 **Am** Americium 95	247 **Cm** Curium 96	247 **Bk** Berkelium 97	251 **Cf** Californium 98	252 **Es** Einsteinium 99	257 **Fm** Fermium 100	258 **Md** Mendelevium 101	259 **No** Nobelium 102	260 **Lr** Lawrencium 103

Key

a **X** b

a = relative atomic mass
X = atomic symbol
b = atomic (proton) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.)

APPENDIX 8: GLOSSARY OF TERMS

It is hoped that the glossary will prove helpful to candidates as a guide i.e., it is neither exhaustive nor definitive. The glossary has been deliberately kept brief with respect to the number of terms included but also to the descriptions of their meanings. Candidates should appreciate that the meaning of a term must depend, in part, on its context.

In all questions, the number of marks allocated is shown on the examination paper, and should be used as a guide by candidates to how much detail to give or time to spend in answering. In describing a process, the mark allocation should guide the candidate about how many steps to include. In explaining why something happens, it guides the candidate on how many reasons to give, or how much detail to give for each reason.

CALCULATE	Used when a numerical answer is required. In general, working should be shown, especially where two or more steps are involved.
COMPARE	identify/comment on similarities and/or differences
DEDUCE	Used in a similar way to "Predict" except that some supporting statement is required (e.g., reference to a law, principle, or the necessary reasoning is to be included in the answer).
DEFINE	(the term(s) ...) is intended literally, only a formal statement or equivalent paraphrase being required.
DESCRIBE	Requires the candidate to state in words (using diagrams where appropriate) the main points of the topic. It is often used with reference either to particular phenomena or to particular experiments. In the former instance, the term usually implies that the answer should include reference to (visual) observations associated with the phenomena. In other contexts, describe should be interpreted more generally (i.e., the candidate has greater discretion about the nature and the organisation of the material to be included in the answer). "Describe and explain" may be coupled, as may "State and explain".
DETERMINE	Often implies that the quantity concerned cannot be measured directly but is obtained by calculation, substituting measured or known values of other quantities into a standard formula (e.g., resistance, the formula of an ionic compound).
ESTIMATE	Implies a reasoned order of magnitude statement or calculation of the quantity concerned, making such simplifying assumptions as may be necessary about points of principle and about the values of quantities not otherwise included in the question.
EXPLAIN	May imply reasoning or some reference to theory, depending on the context.
EVALUATE	judge or calculate the quality, importance, amount, or value of something
FIND	Is a general term that may variously be interpreted as "Calculate", "Measure", "Determine", etc.
GIVE	produce an answer from a given source or recall/memory

IDENTIFY	name/select/recognise
INVESTIGATE	Investigate in the syllabus implies that the learners will have planned the experiment themselves before carrying it out, and as a result will be able to use hypotheses to make predictions and so explain the experimental plan, as well as the issues included above.
LIST	Requires a number of points, generally each of one word, with no elaboration. Where a given number of points is specified this should **not** be exceeded.
MEASURE	Implies that the quantity concerned can be directly obtained from a suitable measuring instrument (e.g., length, using a rule, or mass, using a balance).
NAME	to state or simply specify a word(s) of a known specific feature
OUTLINE	Implies brevity (i.e., restricting the answer to giving essentials).
PERFORM	Perform an experiment in the syllabus implies that the learners will gain great benefit from carrying out such an experiment themselves, and as a result will be able to recall and explain the procedures and the associated science knowledge and understanding, demonstrate how to handle and interpret data from the experiment, and draw conclusions.
PLAN	A detailed step by step procedure followed to test a hypothesis or to arrive at a conclusion
PREDICT	Implies that the candidate is not expected to produce the required answer by recall but by making a logical connection between other pieces of information. Such information may be wholly given in the question or may depend on answers extracted in an earlier part of the question. Predict also implies a concise answer with no supporting statement required.
SKETCH	When applied to graph work, implies that the shape and/or position of the curve need only be qualitatively correct, **but** candidates should be aware that, depending on the context, some quantitative aspects may be looked for (e.g., passing through the origin, having an intercept). In diagrams, sketch implies that simple, freehand drawing is acceptable; nevertheless, care should be taken over proportions and the clear exposition of important details.
STATE	Implies a concise answer with little or no supporting argument (e.g., a numerical answer that can readily be obtained 'by inspection').
SUGGEST	Used in two main contexts (either to imply that there is no unique answer, or to imply that candidates are expected to apply their general knowledge to a 'novel' situation, one that may be formally 'not in the syllabus').

APPENDIX 9: REFERENCE MATERIALS

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